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Research Paper

Gcs-Unet: A lightweight attention network for coaxial melt pool monitoring in laser powder bed fusion

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ABSTRACT

In laser powder bed fusion (L-PBF), coaxial melt pool monitoring methods based on spontaneous radiation often miss low-radiation regions such as the trailing edge, resulting in incomplete information. Additionally, traditional image processing techniques like threshold segmentation lack robustness under complex backgrounds caused by auxiliary lighting, limiting their effectiveness for real-time applications. To address these challenges, a new coaxial melt pool monitoring system was developed, providing clearer and more comprehensive images that capture both geometry and texture. Building on this foundation, an attention-enhanced deep learning network, Gcs-Unet, was proposed to enable robust semantic segmentation under complex conditions. The proposed model achieved an inference time of 6.75 ms while maintaining high performance (99.5 % accuracy, 87.6 % Dice, 86.2 % mIoU) and reducing parameters by 42.91 %, meeting real-time deployment requirements. Furthermore, it was found that scanning speed significantly influences melt pool behavior, with a 33 % speed increase resulting in a 37.45 % rise in the variation of the high-temperature zone's center. These results provide strong support for process optimization and melt pool analysis in L-PBF.

1. Introduction

Laser powder bed fusion (L-PBF), as an advanced additive manufacturing technology, is widely adopted in aerospace, medical, and automotive fields due to its high precision, efficiency, and design freedom, while significantly reducing material waste and enhancing resource utilization [1–3]. In the L-PBF process, the melt pool, as the smallest forming unit, has a significant impact on the quality of the final component due to its geometric characteristics and dynamic behavior. The shape, size, and temperature distribution of the melt pool directly determine the mechanical properties and surface quality of the finished product [4]. Therefore, real-time monitoring of the melt pool's dynamic behavior, ensuring its stability and consistency, is key to achieving high-quality L-PBF manufacturing [5].

The L-PBF process involves multiple physical interactions, including the interaction between the laser and the material, the formation and solidification of the melt pool, heat conduction, and phase transitions of the material [6]. During the processing, the melt pool generates a large amount of acoustic, optical, and thermal signals. Depending on the type of signals, researchers have employed various sensor technologies to monitor the dynamic behavior of the melt pool, including infrared cameras [7], acoustic emission signals [8], photodiodes [9], and high-speed cameras [10]. The formation and solidification of the melt pool is an extremely rapid and complex process, with cooling rates typically ranging from 10^5 to 10^7 K/s, falling within the category of rapid cooling [11]. Such a high cooling rate causes the melt pool to exhibit unique microstructures and performance characteristics during solidification. It is worth noting that defects in the part are formed precisely during this critical period. To monitor this dynamic process in real-time, high-speed cameras, with their unique intuitiveness and high frame rates, have become the most commonly chosen monitoring tool by researchers [12–14]. Based on the camera's placement position, melt pool monitoring can be divided into two types: coaxial monitoring and off-axis monitoring.

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To obtain complete and precise melt pool morphology, a high-speed camera requires a longer focal length and a larger aperture configuration, which will result in a smaller monitoring field of view [15,16]. Additionally, in an off-axis layout, the camera lens forms a certain angle with the processing plane, inevitably leading to geometric distortion in the captured melt pool images, making it difficult to fully and accurately represent the actual melt pool conditions [17]. Compared to the off - axis layout, the coaxial layout has remarkable advantages. In this layout, the camera and processing beam are aligned, eliminating geometric distortion from angular problems. This alignment ensures more precise monitoring results. Additionally, the monitored images can track the processing beam's movement, enabling real - time monitoring of the entire process. This setup not only ensures image quality but also allows for full - scale processing monitoring. It effectively overcomes off - axis layout limitations in field of view and image precision. Kolb et al. [18] used a CMOS camera with a sampling rate of 15 kHz to obtain coaxial images of the L-PBF process and investigated the correlation between melt pool monitoring signals and process parameters. Gaikwad et al. [19] utilized two high-speed cameras of different wavelengths to build a melt pool morphology and temperature monitoring system, combining melt pool shape and spatter characteristics. Furthermore, using the colorimetric temperature measurement method, the system achieved temperature distribution measurement of the melt pool. Zhou et al. [20] employed a coaxial monitoring system combining a high-speed camera and a photodetector to capture melt pool photographs. The study revealed that as the laser energy density increased, both the radiation intensity and temperature of the melt pool significantly improved. Although the coaxial layout increases the manufacturing cost of the printing equipment, the monitoring precision it provides is a key advantage that makes it the preferred choice for this study.

However, current coaxial monitoring methods have a key limitation: when extracting geometric features, researchers primarily rely on the spontaneous radiation of the melt pool to capture its images. As shown in Fig. 1(a), this approach only highlights the high-radiation areas of the melt pool, while overlooking critical information from the tail and other low-radiation regions. Related research indicates that the melt pool images are not always confined to the highlighted regions at the center of the image, which prevents the effective capture of complete melt pool information [21–25]. Moreover, research by Xu et al. [26] indicates that the textural features of the melt pool play a significant role in monitoring. Relying solely on grayscale images for monitoring can result in the loss of critical texture information, further limiting the accuracy and comprehensiveness of the monitoring. Relying solely on high-contrast, binary-like images of the melt pool for monitoring can result in the loss of critical texture information, thereby further limiting the accuracy and comprehensiveness of the monitoring. When extracting features from such melt pool images, threshold segmentation is a commonly used technique; however, it has certain limitations.

Although the threshold segmentation method is simple to operate, it performs poorly in terms of precision and robustness for melt pool segmentation [27]. Research by Fang et al. [28] shows that bright spatters connected to the melt pool can be mistakenly regarded as part of it when using threshold segmentation algorithms to segment melt pool images. It is undeniable that for melt pool images captured using spontaneous radiation methods, the high contrast and relatively simple background allow threshold segmentation to effectively extract the geometric features of the melt pool. However, when auxiliary light sources are introduced to obtain more comprehensive information about the molten pool, the background noise and interference increase, resulting in greater image complexity (as shown in Fig. 1(b)). This renders the threshold segmentation method incapable of accurately extracting the geometric features of the melt pool. In addition, threshold segmentation methods lack the capacity for learning and are unable to effectively handle large-scale datasets, making them inadequate for real-time online monitoring requirements in industrial production. At the same time, the high frame rates of high-speed cameras generate a large amount of data, which also poses challenges for traditional algorithms.

As computer technology advances rapidly, image-processing algorithms, particularly those based on deep learning, are providing new solutions for melt pool monitoring [29–31]. Semantic segmentation, as a crucial deep learning technique, can accurately segment the target areas within an image. Unlike traditional algorithms, it can effectively handle images with complex backgrounds and offers greater robustness and accuracy. Xiao et al. [32] constructed a semantic segmentation network to perform denoising and morphological prediction of the melt pool images. Guo et al. [33] used the FCN - Resnet50 fully convolutional neural network for melt pool image segmentation, achieving a 96.7 % validation accuracy. The network shows excellent segmentation and high robustness. Asadi et al. [34] successfully performed segmentation of melt pool objects using a CNN-based model, which enables the prediction of key parameters such as melt pool area, height, and width. Although CNN-based deep learning algorithms have been widely applied in melt pool image processing, there is still a need for optimization to significantly reduce computational complexity while maintaining high recognition accuracy.

To address the incomplete extraction of melt pool image information and the limitations of image processing algorithms in current coaxial monitoring, this study has developed a new coaxial monitoring system, incorporating auxiliary laser illumination to obtain clearer and more comprehensive melt pool images. By comparing the images from Gaikwad et al. [19] (shown in Fig. 1(a)) and Mazzoleni et al. [23] (shown in Fig. 1(b)), it can be observed that while Mazzoleni et al. [23] utilized auxiliary light sources to slightly enhance the clarity of the melt pool image, it still does not achieve the desired level of clarity. In contrast, the images captured by the constructed monitoring system (shown in Fig. 1 (c)) exhibit higher clarity and completeness, allowing for better capture of melt pool features. This improvement significantly enhances the

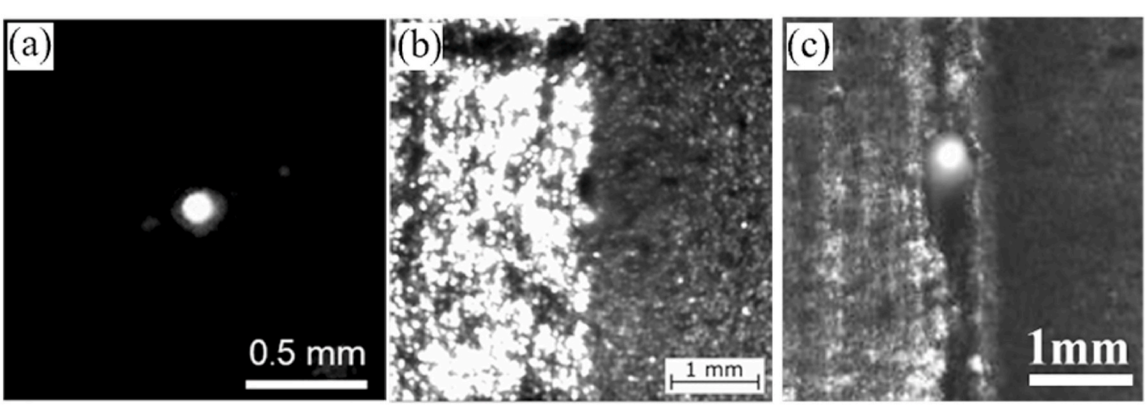

Fig. 1. (a) Typical melt pool image [19]; (b) Melt pool image with an auxiliary light source [23]. (c) The melt pool images collected by our monitoring equipment.

accuracy and reliability of melt pool monitoring. Additionally, this study introduces a Gcs-Unet network that incorporates an attention mechanism and improves upsampling methods, thereby reducing computational parameters while ensuring accuracy.

The original contributions of this research include: (i) a high-speed camera was employed in an on-axis layout to monitor the melt pool images. Compared to current monitoring devices, the developed system can obtain clearer and more accurate melt pool images. (ii) An improved end-to-end convolutional neural network, Gcs-Unet, is presented. This approach significantly reduces the number of model parameters while enhancing accuracy, thereby improving computational efficiency. (iii) the variation laws of melt pool characteristics under different process parameters were explored.

2. Methods

2.1. Experimental setup and parameters

This research utilises a high-speed imaging acquisition system grounded in coaxial machine vision technology to observe the melt pool during the L-PBF fabrication process. Note that the L-PBF system employed in this research was independently developed by our research group. Fig. 2(a) presents a schematic diagram of the monitoring system. The system features a fibre laser with a peak output power of 500W, a central wavelength of 1080 nm (MFSC-500W, Max Photonics), and a concentrated spot diameter of 50 μm. The monitoring system employs a high-speed camera (Revealer, Hefei Agile Device Co., Ltd., Hefei, China) operating at 2000 fps, achieving a spatial resolution of 21.5 μm/pixel. To enhance imaging contrast and enable more accurate extraction of melt pool features, an auxiliary laser with a central wavelength of 808 nm is used to illuminate the processing area. A narrow-band 808 nm optical filter is integrated into the optical path to ensure selective capture of the reflected signal from the melt pool surface, as the 808 nm laser (FC-W-808nm-150W-DF61528, Changchun New Industries Optoelectronics Tech. Co., Ltd, China) serves as dispersed illumination. This setup allows for the clear visualization of melt pool features as small as 100 μm. Fig. 2(b) shows a physical diagram of the monitoring system. Given that various scanning strategies influence the thermal-mechanical behaviour of components, a bidirectional scanning technique is employed for printing [35]. Fig. 2(c) depicts the scanning approach and the orientation of the sample during construction.

Given the experimental expenses, 316L stainless steel powder was employed for printing, including an average particle size of 43 μm and a diameter distribution between 30 and 75 μm. SEM images of 316L powder are shown in Fig. 3. The substrate material consists of 316L stainless steel. The laser power varied from 200W to 350W and the scanning speed fluctuated between 60 and 100 mm/s. The hatch spacing was maintained at 50 μm, the powder bed thickness measured 50 μm, and a total of 5 layers were printed in the experiments. Before the experiment begins, preheat the substrate to 200 °C and start printing. The dimensions of the printed sample were 10 mm × 10 mm, with a 3 mm interval between neighbouring samples. To thoroughly assess the impact of different factors on the test specimens, it is standard practice to quantify using either the laser volume energy density or the laser surface energy density [36]. This study utilises the laser volume energy density, as defined in Equation (1), with parameter configurations detailed in Table 1. The matrix includes nine groups of process parameters, designed to systematically span a broad energy density range (1000.0–2333.3 J/mm^3) by varying laser power (200–350 W) and scanning speed (60–100 mm/s), as presented in Table 1.

$$E_V = \frac{P}{v \times h \times s} \quad (1)$$

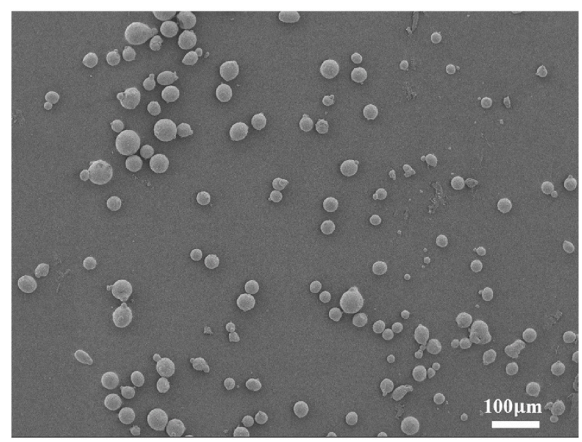

Fig. 3. SEM images of 316L powder.

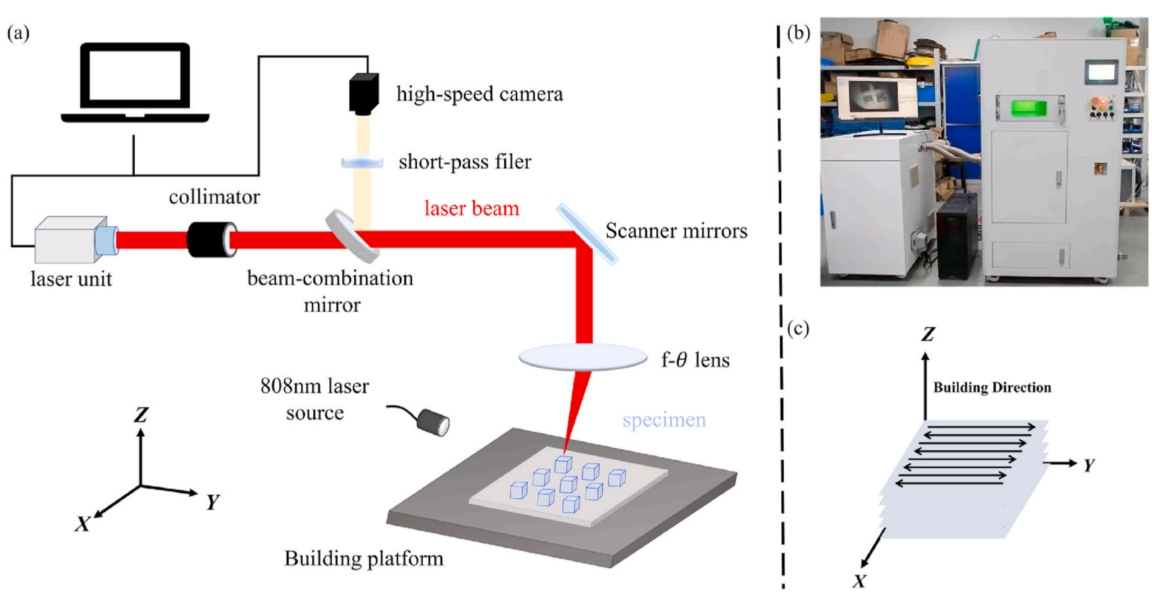

Fig. 2. (a) Schematic diagram of the coaxial monitoring system; (b) Photograph of the experimental equipment; (c) Schematic diagram of the scanning strategy and build direction.

Table 1
Experimental parameters.

Sample Number	P (W)	v (mm/s)	EV (J/mm^3)
1	200	60	1333.3
2	250	60	1666.7
3	300	60	2000.0
4	350	60	2333.3
5	200	80	1000.0
6	250	80	1250.0
7	300	80	1500.0
8	350	80	1750.0
9	350	100	1400.0

E_V is the laser volume energy density (J/mm^3), P is the laser power (W), h is the hatch spacing (mm), v is the scanning speed (mm/s), s is the layer thickness (mm).

In the field of AM techniques, maintaining high-quality consistency, dimensional accuracy, and process repeatability remains a critical challenge [37,38]. This challenge fundamentally stems from the high sensitivity of process parameters—subtle variations in parameters such as laser power and scanning speed can significantly affect melt pool dynamics and forming quality [39]. To this end, this study deliberately designed these nine groups of unoptimized process parameters to simulate the common parameter fluctuation scenarios in actual L-PBF production.

Experimental results show that most parameter combinations resulted in melt pool instability or forming defects, making reliable feature extraction difficult. Only Samples 4 and 8 formed stable and clearly defined melt pools under the same laser power but with different scanning speeds. The single-variable characteristic provided an ideal comparison for precisely analyzing the influence of this parameter on melt pool stability. In light of this, this study focused on these two groups of samples for subsequent dataset construction and in-depth analysis, while other samples with substandard quality were not included in the detailed discussion.

2.2. Image data acquisition and preprocessing

The monitoring system captures 1920 × 1080 pixel images in RGB three-channel format. Their large size can introduce significant noise, hindering model training and challenging the processing capacity of both the model and equipment. To align with the model's input requirements, we preprocessed the images by cropping the region of interest (ROI) around the melt pool, focusing on the melt pool area to obtain 160 × 160 pixel images.

During the data collection phase, each sample was divided into 5 parts, and images were captured for adjacent scanning paths in each part. Consequently, each sample corresponded to 5 standardized datasets, resulting in a total of 10 datasets containing 2283 images. The specific data collection process is illustrated in Fig. 4(a). These images were split into training and validation sets at a ratio of 9:1. Image annotation was performed using Labelme 5.4.2 software [40], with annotation categories covering background, melt pool, and high-temperature area of the melt pool.

It should be noted that the annotation protocol adopted in this study follows the approaches of Zheng et al. [7] and Mao et al. [31]. Specifically, the melt pool region is delineated by the solidus line, while the high-temperature area of the melt pool is defined based on the distinct contrast features observed in the images. The detailed annotation results are illustrated in Fig. 4(b).

2.3. Gcs-Unet architecture

The proposed Gcs-Unet enhances the Unet architecture, as depicted in Fig. 4. Gcs-Unet utilises bilinear interpolation for upsampling instead of transposed convolutions, while the downsampling method remains

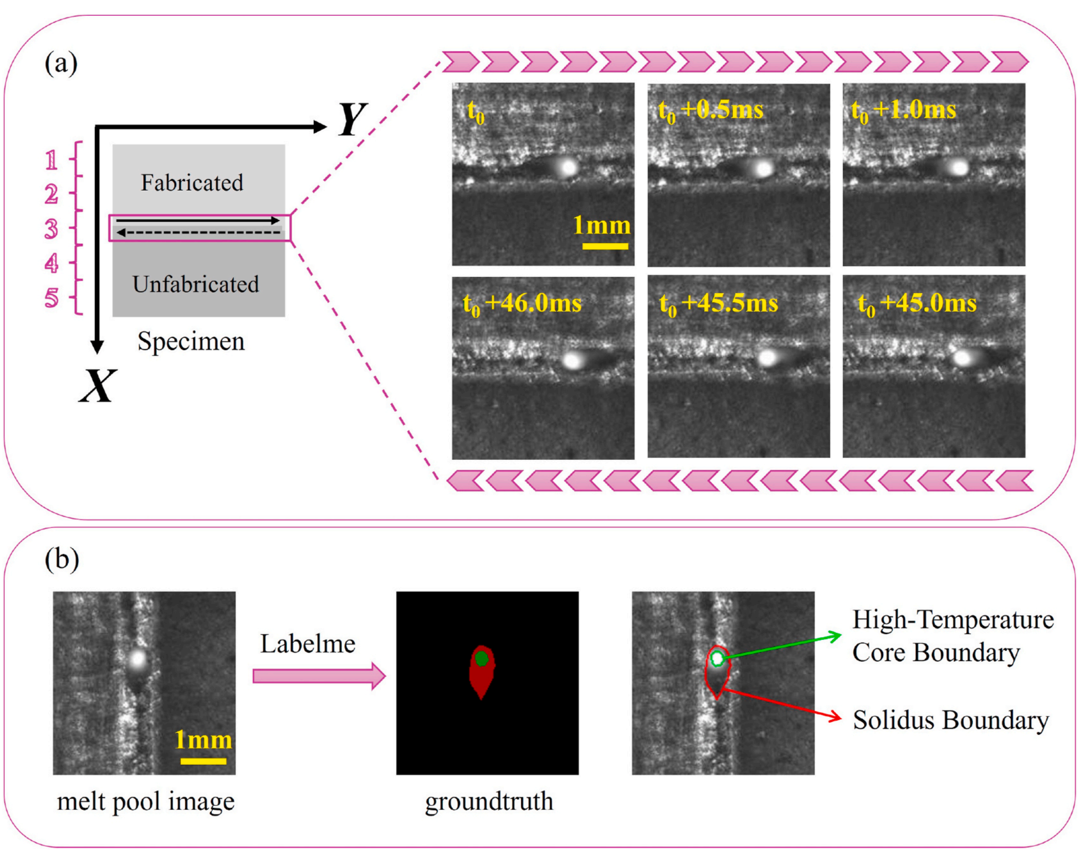

Fig. 4. (a)Data Collection; (b) Melt pool region delineation.

consistent with the original Unet [41]. Although the ROI region is captured in the melt pool image, the melt pool region is still a typical small target. To effectively segment the melt pool, the bilinear interpolation upsampling method is employed to enhance the smoothness of feature maps, thereby better preserving image details [42]. This is crucial for detecting smaller melt pool targets and improving the model's detection accuracy and robustness. To further refine the melt pool region extraction, an attention mechanism is integrated into the network. It enhances the model's focus on the target region, improves boundary sensitivity, and reduces background noise interference [43]. Consequently, a Gated Channel and Spatial Attention Mechanism module (GCSAM) is developed, as illustrated in Fig. 5, to boost feature extraction and improve melt pool segmentation.

Fig. 6(a) illustrates the overall structure of the GCSAM Block designed in this study. The input feature map first undergoes a 1 × 1 convolution operation, followed by sequential inputs to the channel attention and spatial attention modules, and concludes with another 1 × 1 convolution operation. The computation of the feature map in Fig. 6(a) can be defined by Equations (2) and (3). Given a feature map $\boldsymbol{M} \in R^{C\times H\times W}$ as input, the channel attention module $CA(\cdot)$ outputs channel weights, which are then element-wise multiplied with the input feature $\boldsymbol{M}$ to obtain the refined feature $\boldsymbol{M_C} \in R^{C\times H\times W}$. Where H represents the height of the feature map, W is the width, and C is the number of channels. The spatial attention module processes $SA(\cdot)$ the channel attention map $\boldsymbol{M_S}$, $\boldsymbol{M_S} \in R^{C\times H\times W}$.

$$\boldsymbol{M_C} = CA(\boldsymbol{M}) \oplus \boldsymbol{M} \tag{2}$$

$$\boldsymbol{M_S} = SA(\boldsymbol{M_C}) \otimes \boldsymbol{M_C} \tag{3}$$

The channel attention module incorporates three trainable parameters, α, β, and γ, and $l_1 -$ norm employs to process the feature map across channels. Inspired by ResNet network [44], a gating mechanism based on residual network is designed to improve the robustness of deep networks, as shown in Fig. 6(b). The embedding weights $\alpha = [\alpha_1, \cdots, \alpha_C]$ are initialized to 1. The parameter α aggregates information within each channel to prevent local semantic ambiguities. Channel normalization establishes a competitive relationship between channels, incentivizing those with higher weights and suppressing those with lower weights. The gating weight γ and the bias β facilitate both competitive and cooperative relationships among channels [45]. When the activation function outputs a positive value, it establishes a competitive relationship, while a negative output fosters a cooperative relationship. The following processes can be defined by Equations (4)–(6). Let $\boldsymbol{M} = [m_1, m_2, \dots, m_C]$, $m_c = \left[m_c^{i,j}\right]_{H\times W} \in R^{H\times W}$, c ∈ {1,2, …C}. m_c denotes the two-dimensional tensor of the feature map $\boldsymbol{M}$ at channel c.

$$\overline{m}_C = \alpha_C \left\{ \left[\sum_{i=1}^{H} \sum_{j=1}^{W} \left| m_c^{i,j} \right| + \varepsilon \right] \right\} \tag{4}$$

$$\widehat{m}_c = \frac{\sqrt{C}\overline{m}_c}{\left[\left(\sum_{c=1}^{C} \left| \overline{m}_c \right| \right) + \varepsilon \right]} \tag{5}$$

$$\widetilde{m}_c = m_c + m_c \tanh(\gamma_c \widehat{m}_c + \beta_c) \tag{6}$$

In the aforementioned equations, ε is a small constant to prevent derivative issues at zero points, with a value of 1e-5 used in this paper. *tanh* () represents the neuron activation function.

Fig. 6(c) shows the schematic diagram of the spatial attention module in the GCSAM module. The spatial attention module adopts depth-wise convolution to efficiently capture spatial dependencies while reducing computation [46]. A multi-scale structure is further introduced to enhance spatial feature extraction. Finally, a 1 × 1 convolution is applied at the end of the module to mix channel information and generate a more precise attention map. The calculation method for spatial attention is shown in Equation (7):

$$SA(M_C) = Conv_{1\times1}\left(\sum_{0}^{3} Scale_i(DW - Conv(M_C)) \right) \tag{7}$$

In the equation, *Dw-Conv* represents depthwise convolution, and Scalei, where i∈{0,1,2,3}, represents the i-th branch.

2.4. Model optimization and evaluation criteria

The training was conducted on a computer system running Ubuntu 18.04, equipped with an NVIDIA RTX 3090 GPU. The code was executed in an environment with PyTorch 1.9 and Python 3.8.5. The training parameters are summarized in Table 2. Given the imbalance in the dataset with a dominance of background pixels and fewer foreground pixels related to the melt pool, using a combination of Cross-Entropy and DiceCoefficient as the loss function (LCE + LDice) is proposed to

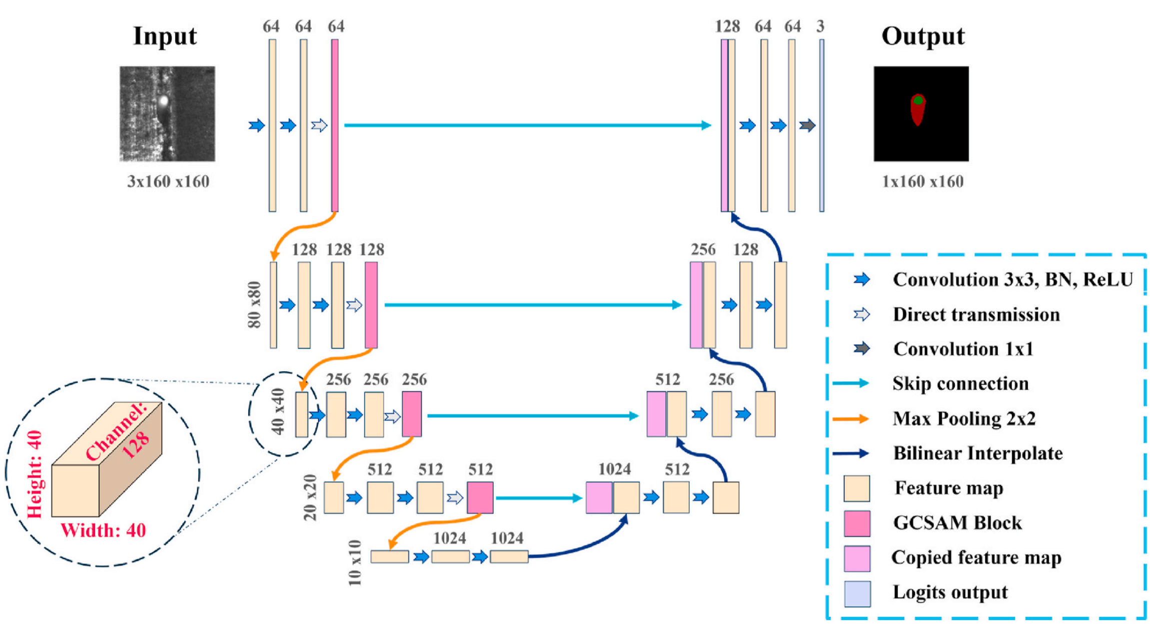

Fig. 5. Gcs-Unet architecture.

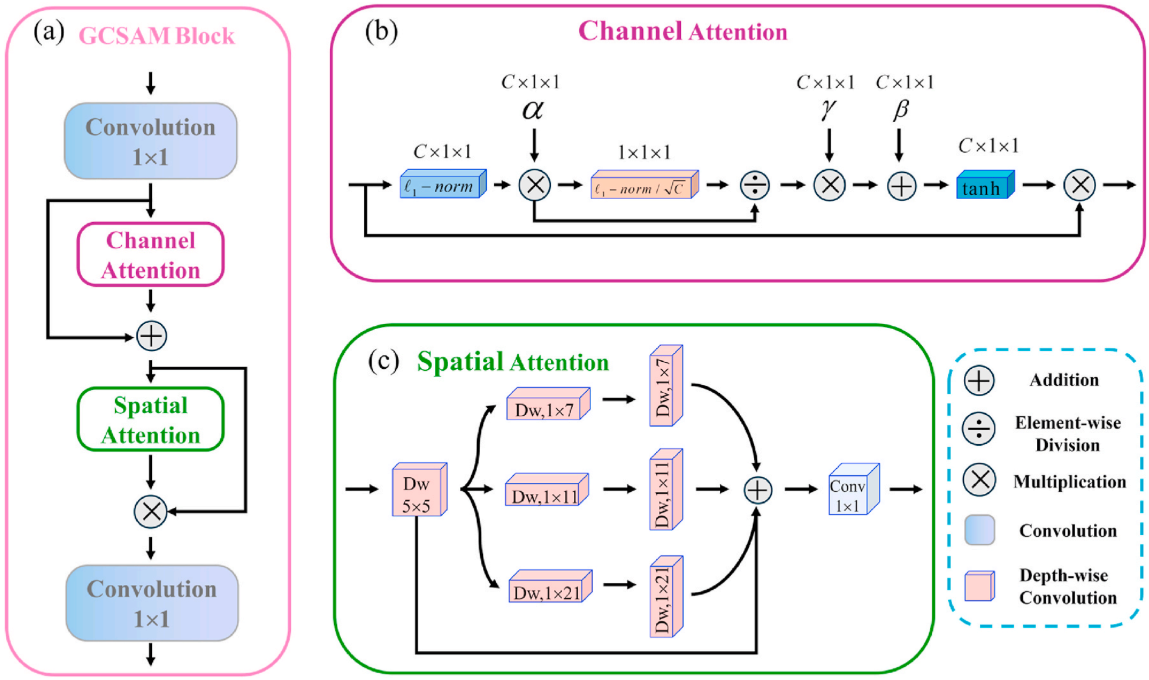

Fig. 6. (a) Gated channel and spatial attention mechanism block (GCSAM Block); (b) channel attention; (c) spatial attention.

Table 2
Training parameters.

Parameter	Value
Epoch	100
Batch size	64
Loss function	Dice loss + cross-entropy loss
Optimizer	Adam
Learning rate	0.01
Train: Test	9:1

enhance model training. To address the model's tendency to overfit background features due to the small proportion of the melt pool in the image, a combination of cross-entropy and Dice loss (LCE + Ldice) is employed, as defined in Equation (8). This integrated loss enhances the model's focus on the melt pool region and improves segmentation and detection accuracy.

$$L = 1 - \frac{2}{C}\sum_{k=1}^{C}\frac{\sum_{i=1}^{N} y_{k,i}\widehat{y}_{k,i}}{\sum_{i=1}^{N} y_{k,i} + \sum_{i=1}^{N}\widehat{y}_{k,i}} - \frac{1}{CN}\sum_{k=1}^{C}\sum_{i=1}^{N} y_{k,i}\log\left(\widehat{y}_{k,i}\right) \tag{8}$$

Let C represent the number of classes, corresponding to the melt pool and the high-temperature area of the melt pool. Let N denote the number of samples, where $y_{k,i}$ indicates the true value of the i-th sample belonging to class k, and $\widehat{y}_{k,i}$ represents the corresponding predicted value.

In this study, the evaluation metrics employed are Pixel Accuracy (PA), Mean Intersection over Union (mIoU), and Dice Coefficient (Dice). These metrics are defined as follows: PA reflects the proportion of correctly classified pixels against the total pixels, as shown in Equation (9). The mIoU, presented in Equation (10), measures the average ratio of the intersection to the union between predicted and ground-truth regions across all classes. The Dice, given in Equation (11), evaluates the similarity between predicted and actual regions, with higher values indicating better performance.

$$PA = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \tag{9}$$

$$MIoU = \frac{1}{k+1}\sum_{i=0}^{k}\frac{TP}{FN + FP + TP} \times 100\% \tag{10}$$

$$Dice\_Coefficient = \frac{2 \times TP}{2 \times TP + FP + FN} \times 100\% \tag{11}$$

Where k and i are the number of classes, and the i-th class. The terms are defined as follows: TP (True Positive) is the count of samples correctly predicted as positive and actually positive; FP (False Positive) is the count of samples incorrectly predicted as positive but actually negative; TN (True Negative) is the count of samples correctly predicted as negative and actually negative; FN (False Negative) is the count of samples incorrectly predicted as negative but actually positive.

3. Results and discussion

3.1. Assessment of model segmentation results

3.1.1. Comparison with threshold segmentation

To highlight the limitation of threshold segmentation methods in achieving effective segmentation for melt pool images with complex backgrounds, this paper conducts comparative validation of the extraction performance between the Gcs-Unet network and threshold-based methods (including global thresholding, iterative thresholding, and the Otsu algorithm), with results presented in Fig. 7. Among them, Fig. 7(a) shows the original melt pool images randomly selected from the validation set, and Fig. 7(b) displays manually annotated ground truth images. The complete melt pool is subdivided into two parts: the melt pool high-temperature zone and the melt pool main body, corresponding to the green and red regions in Fig. 7(b), respectively. The definitions of these categories are elaborated in Section 2.2.

The results indicate that threshold segmentation methods perform poorly in extracting melt pools captured using active illumination, with significant background interference. Although the three thresholding methods can clearly obtain the melt pool high-temperature zone, their performance in extracting the remaining melt pool areas is subpar. The iterative thresholding (Fig. 7(d)) and Otsu algorithm (Fig. 7(e)) results show substantial background noise and inferior segmentation

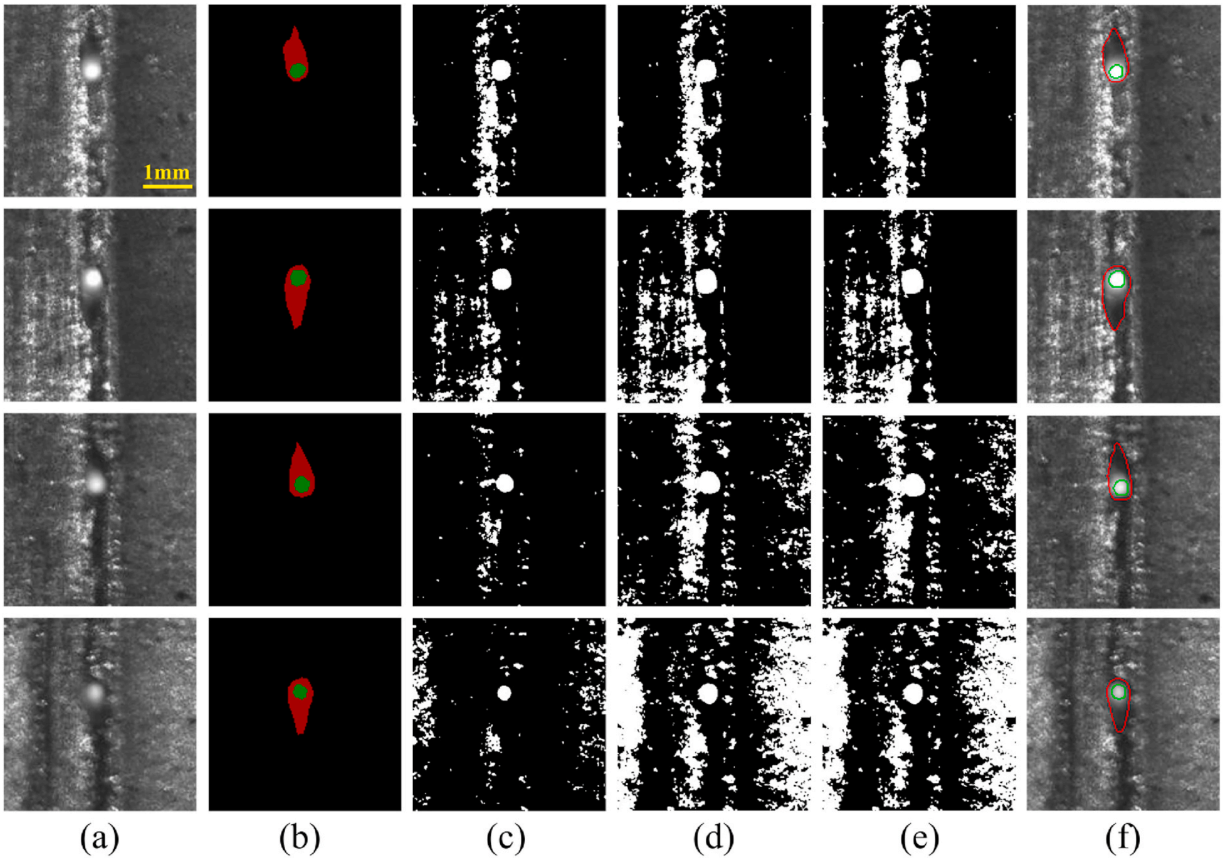

Fig. 7. (a)Input; (b)Groundtruth; (c) Global Thresholding Result; (d) Iterative Thresholding Result; (e) Otsu Thresholding Result; (f) Gcs-Unet result.

refinement compared to global thresholding (Fig. 7(c)). Fig. 7(f) displays the Gcs-Unet segmentation results, with edges enhanced by the Canny operator. Gcs-Unet outperforms the thresholding methods in segmentation accuracy and refinement.

When monitoring melt pools using a coaxial system, passive monitoring, as used by Gaikwad et al. [19], is more common. This method yields melt pool images with only the bright zone and surrounding areas, reducing background interference and allowing thresholding methods to perform well. However, the actual melt pool shape may differ from what is shown in the bright zone due to process parameters and material effects, potentially leading to inaccurate feature extraction.

Despite the prevalence of passive monitoring due to equipment precision and layout considerations, the active monitoring method presented in this article captures more precise melt pool images. The proposed image processing effectively filters out interference, providing accurate melt pool images and a novel monitoring approach.

3.1.2. Comparison of the improved model with the benchmark model

To demonstrate the performance of the improved model, a comparison was made between the baseline model and the improved model on the validation set, with the results shown in Fig. 8. Fig. 8(a) indicates that both models have a stable training process and good convergence. After 100 epochs, the pixel accuracy of the models reaches 99.5 %. The models exhibit high accuracy in pixel-level, end-to-end segmentation of images, ensuring the accuracy of melt pool area extraction. Fig. 8(c) and (d) display the changes in Dice coefficient and mIoU for both models during the training process. The Dice coefficient of the baseline model increased from 86.8 % to 87.6 %, while the mIou of the improved model increased from 85.5 % to 86.2 %. The improved model surpasses the baseline model in terms of the Dice coefficient, and consistent results are also obtained at the level of mIoU.

3.1.3. Comparison with other semantic segmentation models

The Gcs-Unet model's competitiveness was assessed by comparing it with other leading semantic segmentation models, including FCN [47] with Resnet50 as the backbone (FCN-Resnet50), DeepLabV3 [48] utilising Resnet50 as the backbone (FCN-Resnet50), Lraspp [49] with MobileNet as the backbone, Unet, and O-Net [50].

The results of this comparative study (Table 3) demonstrate that the proposed Gcs-Unet achieved the highest values across three key metrics: mIoU (86.2 %), Dice coefficient (0.876), and PA (99.5 %). This outcome highlights the trade-off that must be made between segmentation performance and model parameter volume. Overall, more complex and deeper network architectures, such as Unet, DeepLabV3, and FCN, along with their respective backbone networks, exhibit superior performance. In contrast, lightweight networks like Lraspp, despite having fewer parameters, fall short in terms of accuracy.

It is noteworthy that O-net delivers better performance than DeepLabV3 and Lraspp while maintaining the smallest model size and computational complexity, highlighting its efficiency. However, as shown in Fig. 9(c), O-net still exhibits shortcomings in segmenting melt pool images. This may be attributed to the limited size and relatively simple features of our dataset, which could also affect the performance of other lightweight models such as DeepLabV3 and Lraspp.

Fig. 9 further illustrates the performance differences among models: O-net lacks fine detail processing, DeepLabV3 performs poorly on blurry images, and Lraspp fails to capture high-temperature regions effectively. FCN performs relatively well on blurry melt pool images, while the original Unet achieves good overall results but still leaves room for improvement in fine structures. In contrast, the proposed Gcs-Unet demonstrates superior segmentation on both clear and blurry melt pool images, confirming its robustness and potential for practical application.

The superior performance of Gcs-Unet benefits from the integration

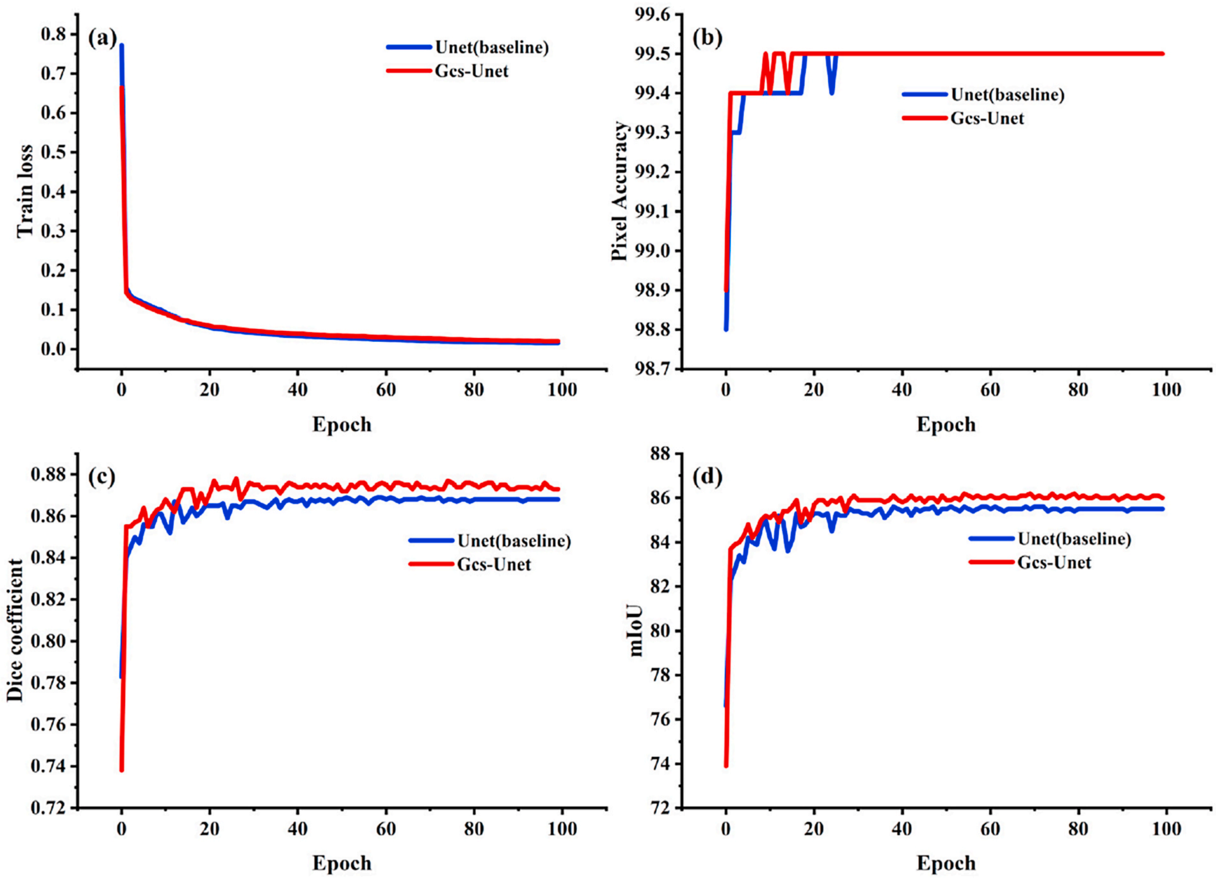

Fig. 8. Training Results.(a)Train loss; (b) Pixel Accuracy; (c) Dice Coefficient; (d) mean Intersection over Union.

Table 3
Performance of different semantic segmentation models on the validation.

model	Backbone	mIou (%)	PA (%)	Dice (%)	Number of parameters(M)
Unet [41]	–	85.5	99.5	86.8	31.04
FCN [47]	Resnet50	84.0	99.4	85.5	32.95
DeepLabV3 [48]	Resnet50	82.2	99.3	83.4	39.64
Lraspp [49]	MobileNetV3	80.8	99.3	81.9	3.22
O-net [50]	–	83.8	99.4	85.3	0.48
Gcs-Unet	–	**86.2**	**99.5**	**87.6**	**17.72**

of a depthwise separable convolution-based attention mechanism, which enhances segmentation accuracy while significantly reducing parameter count. Moreover, replacing the conventional upsampling operations with bilinear interpolation further reduces computational overhead, contributing to the model's lightweight and efficient design.

In summary, deeper and more complex networks tend to perform better in segmentation tasks, while lightweight architectures often suffer from detail loss. Gcs-Unet strikes a balance between accuracy and robustness, with a reduced parameter footprint, making it highly suitable for melt pool image segmentation in practical applications.

3.1.4. Ablation study

To verify the effectiveness of the improved model and to address the demands of real-time online monitoring in L-PBF powder bed processes, this paper conducted an ablation study using Unet as the baseline model. The main experiments involved the following networks: (1)Unet network with transposed convolution for upsampling (baseline model); (2) Unet + GCSAM, which incorporates the GCSAM module into the Unet; (3) Unet + bilinear interpolation (Unet + BI), which changes the upsampling part to bilinear interpolation without incorporating the GCSAM module; (4) Gcs-Unet, which adds the GCSAM module and replaces the transposed convolution in the decoding part of the network with bilinear interpolation.

The results of the ablation study are shown in Table 4. To ensure the reliability and statistical robustness of the findings, each module involved in the ablation study was subjected to 5 independent experimental trials. The reported results in Table 4 represent the average values derived from these repeated experiments, thereby mitigating the potential impact of random variations on the outcome. These results demonstrate clear trade-offs between model accuracy, parameter count, and real-time inference speed, which are critical for practical deployment in online monitoring applications. Introducing the GCSAM module led to improvements in model performance, with mIoU increasing from 85.42 $\pm$ 0.16 % to 85.88 $\pm$ 0.08 % and Dice coefficient from 86.80 $\pm$ 0.07 % to 87.50 $\pm$ 0.25 %. However, this improvement came at the cost of an increased number of parameters (from 31.04M to 31.50M) and a substantial rise in inference time (from 4.10 ms to 7.52 ms per frame), as the attention mechanism, though parameter-efficient, introduces additional computation during inference.

Replacing transposed convolution with bilinear interpolation (Unet + BI) not only maintained performance gains (mIoU also increased to 85.74 $\pm$ 0.15 %) but, more importantly, drastically reduced the parameter count from 31.04M to 17.26M, a reduction of 44.39 %. This architectural change also resulted in the lowest inference time among all methods (3.20 ms per frame), highlighting the advantage of bilinear interpolation for real-time applications due to its non-parameter nature and computational simplicity.

The final improved model, Gcs-Unet, which integrates both GCSAM and bilinear interpolation, achieved the highest mIoU (86.12 $\pm$ 0.08 %) and Dice coefficient (87.54 $\pm$ 0.13 %), while keeping the parameter

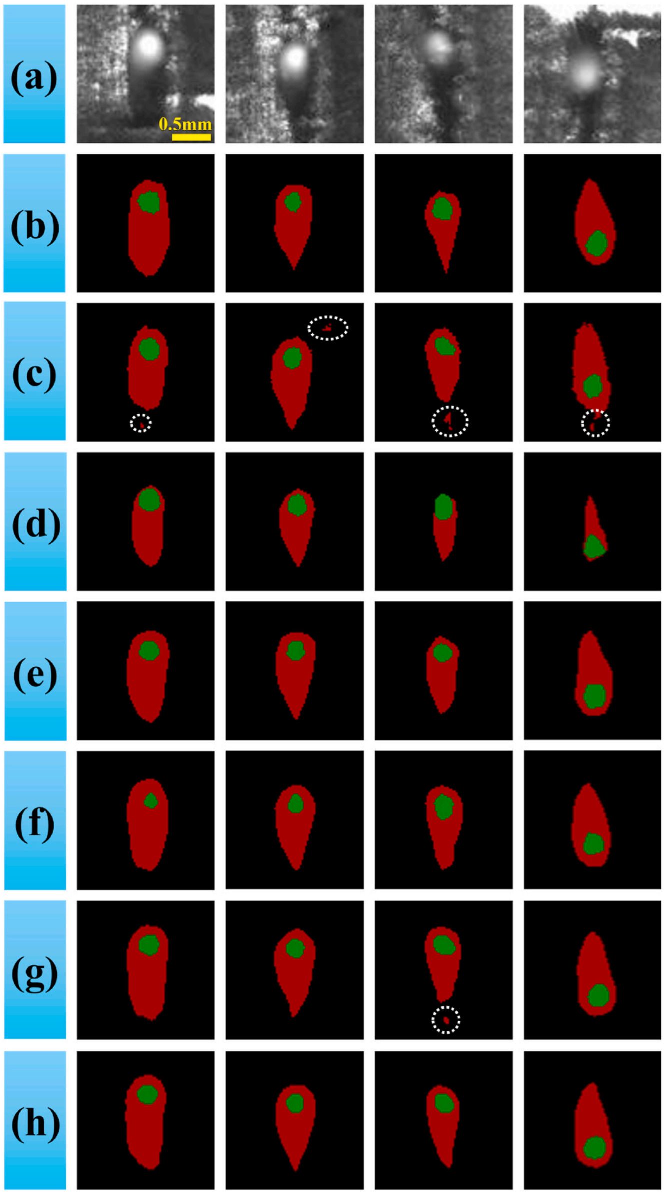

Fig. 9. Segmentation results of different semantic segmentation models on the Validation. (a) melt pool image; (b) Groundtruth; (c) O-net; (d) DeepLabV3 - ResNet50; (e) FCN-ResNet50; (f) Lraspp; (g) Unet; (h) Proposed (Gcs-Unet). The white circle is to highlight the location of the prediction error.

count at 17.72M—a 42.91 % reduction compared to the baseline Unet. Although the inference time (6.75 ms per frame) is slightly higher than that of Unet + BI, it is still significantly faster than Unet + GCSAM and remains well-suited for real-time online monitoring scenarios.

These results demonstrate consistent superiority in preliminary validations. For the ablation study (Table 4), five independent trials per module yielded statistically averaged metrics. Slight deviations between initial and averaged data originate from training stochasticity and experimental variations – inherent to deep learning-based segmentation. Collectively, Gcs-Unet achieves an optimal accuracy-efficiency-inference latency balance, enabling real-time L-PBF melt pool feature extraction. This confirms its readiness for industrial deployment.

3.2. Melt pool geometry prediction

This study established a dynamic feature extraction system for melt pools by combining the Gcs-Unet network with edge detection algorithms. The main monitoring parameters include: one category of directly measured parameters, such as melt pool area and high-temperature zone area, which are calculated based on the number of pixels in the segmented region and the squared pixel-to-length conversion factor (21.5 μm/pixel); and another category of indirectly deduced parameters, including melt pool long axis length, short axis width, and centroid coordinates (based on the minimum bounding rectangle algorithm). A dual-channel feature extraction strategy was adopted to simultaneously capture the overall morphology and dynamic characteristics of the high-temperature core zone of the melt pool. By converting complex geometric features into quantifiable engineering parameters using the minimum bounding rectangle, the system meets the requirements of closed-loop control for high-frequency data acquisition.

Fig. 10 shows the geometric prediction of the melt pool for Sample 4’s second set of data, including melt pool area, high-temperature zone area, melt pool length, melt pool width, high-temperature zone length, and high-temperature zone width. As shown in Fig. 10, at the laser path turning point (Frame 132), the melt pool area, long axis, and short axis all exhibit “V-shaped” mutation characteristics. Before the turning point, the feature values continuously increase and then tend to stabilize; near the turning point, they begin to decrease, reaching minimum values at the turning point. After leaving the turning point, the feature values quickly increase to a stable region. The overall trends of the melt pool parameter changes and the high-temperature zone parameter changes are similar, but the overall melt pool changes in terms of area and length are more significant than those of the high-temperature zone parameters, which is determined by their geometric dimensions.

Surface morphology measurements (Fig. 11) revealed that protrusion defects are generally present in the laser turning regions, with the severity of these defects showing a dependence on process parameters. For Sample 8 (Parameter Group 8), the protrusion height reached 705.70 μm (Fig. 11(b)), exhibiting obvious spheroidization characteristics. For Sample 4 (Parameter Group 4), the protrusion height decreased to 476.01 μm, and the morphology became more planar. The energy density distribution at the turning point undergoes distortion, as the sudden change in the scan mirror’s acceleration at the turning point leads to an extended laser dwell time, resulting in increased local energy input and a tendency for the melt pool area to grow. After leaving the turning point, the energy input becomes uniform, and the changes in melt pool feature parameters also become uniform. However, near the turning point, the laser incidence angle θ increases, and based on Fresnel’s reflection law, the reflection energy loss increases, leading to a decrease in effective melting energy.

To quantitatively evaluate the prediction accuracy of Gcs-Unet for melt pool features, this study adopted a dual-index evaluation system based on mean square error (MSE) and mean relative error (MRE). As shown in Table 5, the model exhibited differentiated precision in predicting the area of the overall MP area and HT area. In terms of absolute

Table 4
Ablation study results.

model	GCSAM	BI	mIou (%)	PA (%)	Dice (%)	Number of Parameters (M)	Inference Time (ms)
Unet	×	×	85.42 ± 0.16	99.36 ± 0.31	86.80 ± 0.07	31.04	4.10
Unet + GCSAM	✓	×	85.88 ± 0.08	99.48 ± 0.04	87.50 ± 0.25	31.50	7.52
Unet + BI	×	✓	85.74 ± 0.15	99.22 ± 0.42	86.96 ± 0.05	17.26	3.20
Gcs-Unet	✓	✓	86.12 ± 0.08	99.46 ± 0.09	87.54 ± 0.13	17.72	6.75

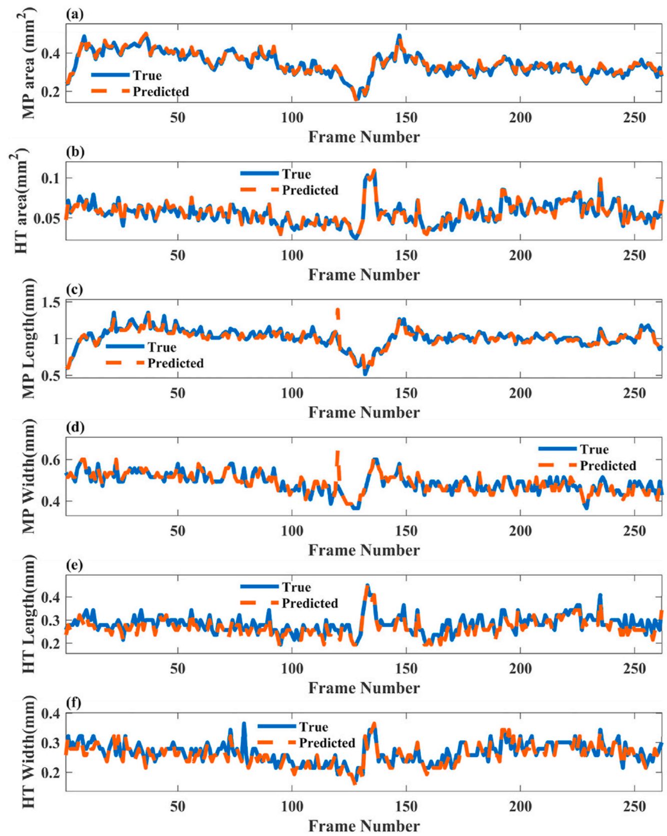

Fig. 10. Data 4_2 Melt Pool Geometry Prediction; (a) Melt pool area (MP area); (b) High-temperature zone area of the melt pool (HT area); (c) MP length; (d) MP width; (e) HT length; (f) HT width.

error, the MSE for MP area and HT area was 8.0×10^{-4} and 1.0×10^{-4}, respectively, indicating similar orders of magnitude in precision. However, in terms of relative error, the MRE for HT area (13.876 %) was significantly higher than that for MP area (6.919 %), with a difference of 2.01 times. This phenomenon reveals the limitation of the traditional MSE metric in evaluating micro-scale features: even when absolute errors are low, relative errors can still be significantly amplified when the physical dimensions of the features are small. The adoption of an MSE-MRE joint evaluation strategy in this study allows for a more comprehensive reflection of the model's prediction stability across macro and micro scales.

The model's predictions for MP length and width showed consistency in MRE. The relative errors for the two parameters were 6.824 % and 6.866 %, respectively, with a difference of only 0.6 %, indicating that the model's sensitivity to scale in different axial directions was highly consistent. However, in terms of MSE, the absolute errors were 0.0226 mm (length) and 0.0068 mm (width), a difference that arises from the larger physical scale of the long axis itself.

These results demonstrate that Gcs-Unet can effectively capture the morphological features of the melt pool, with prediction accuracy that meets the requirements of L-PBF process monitoring. In particular, in the prediction of axial parameters, the collaborative validation of MSE and MRE proves the model's ability to accurately maintain the proportional relationships of geometric features.

3.3. The influence of process parameters on melt pool characteristics

This study analyzes the melt pool characteristics under different scanning speeds as part of the process monitoring and evaluation. As shown in Fig. 12, box plots were used to visualize the distribution of feature parameters, identify and remove outliers, and extract the mean values for each scanning speed. To quantify the positional consistency of the melt pool and its thermal profile, two offset indicators were introduced: MP_distance, defined as the Euclidean distance between the melt pool center (approximated by the center of its bounding box) and the center of the ROI image; and HT_distance, similarly defined for the high-temperature zone. The results show that, under otherwise consistent process conditions, the mean values of melt pool area, length, width, and both offset distances gradually decreased with increasing scanning speed. Similar trends were observed for the high-temperature zone.

To further quantify these observations, a quantitative analysis was conducted by comparing the coefficient of variation (CV values) in two groups of experimental data (Sample 4 at v = 60 mm/s and Sample 8 at v = 80 mm/s), as detailed in Table 6. This analysis systematically evaluates the impact of scanning speed on melt pool dynamics. The findings reveal that changes in scanning speed significantly affect the dynamic stability of the melt pool. Based on the magnitude of CV value changes, the study categorizes melt pool features into four sensitivity levels: extremely high sensitivity features, high sensitivity features, medium sensitivity features, and low sensitivity features. This

Table 5
Validation set melt pool error calculation results.

Feature	MSE	MRE
MP area	0.0008 mm^2	6.919 %
HT area	0.0001 mm^2	13.876 %
MP length	0.0226 mm	6.824 %
MP width	0.0068 mm	6.866 %
HT length	0.0018 mm	9.402 %
HT width	0.0010 mm	9.541 %

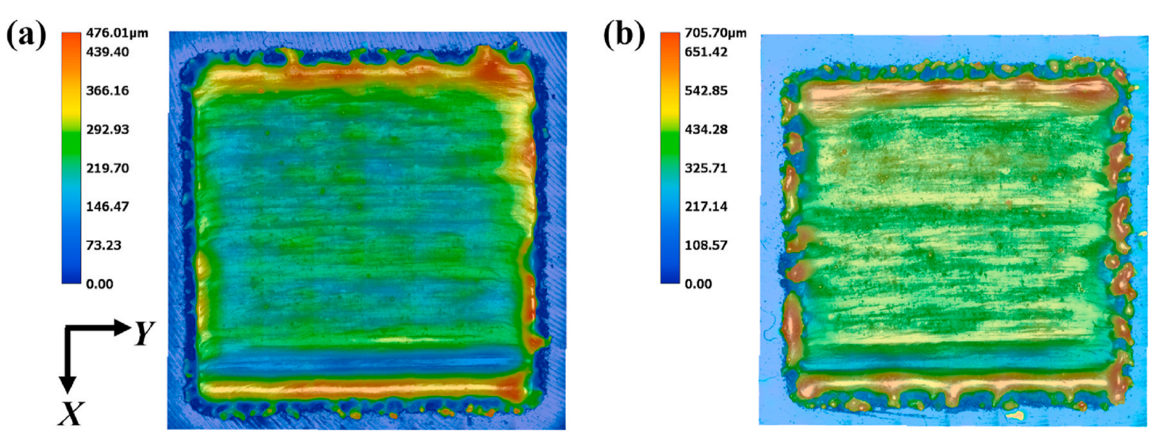

Fig. 11. The Keyence super-depth-of-field result image of the sample surface. (a) Sample 4; (b) Sample 8.

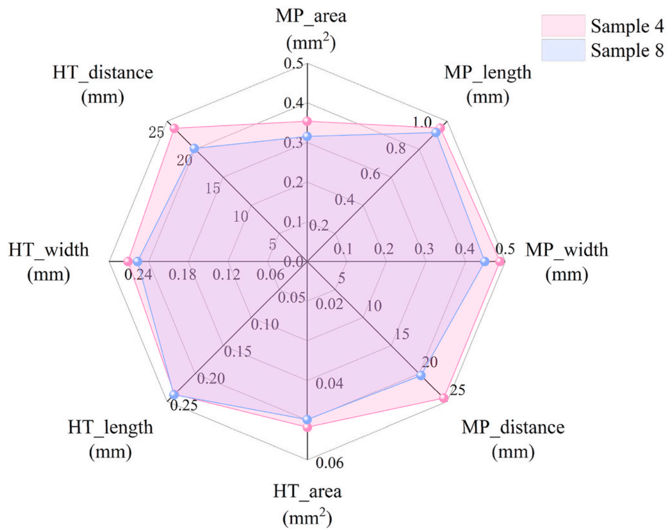

Fig. 12. Comparison of melt pool feature changes between sample 4 and sample 8.

Table 6
Validation set melt pool error calculation results.

Feature	Sample 4 CV ($\times 10^{-2}$)	Sample 8 CV ($\times 10^{-2}$)	ΔCV (%)	level
HT distance	6.53	8.98	37.45	Extremely High
HT length	12.92	16.30	26.17	High
MP area	13.42	16.80	25.13	High
MP distance	58.78	70.29	19.57	Medium
MP width	7.55	8.87	17.55	Medium
MP length	9.04	10.61	17.48	Medium
HT area	21.86	25.43	16.34	Medium
HT width	12.44	13.06	4.94	Low

comprehensive approach combines qualitative and quantitative analyses to provide a deeper understanding of how scanning speed influences melt pool characteristics.

The research revealed that the coefficient of variation (CV) of the geometric center offset of the high-temperature zone (HT dis) increased by 37.45% when the scanning speed was raised by 33%, indicating that it is an extremely sensitive feature. Specifically, the increase in scanning speed significantly intensified the fluctuation of the geometric center. This is primarily due to the shorter laser dwell time at higher speeds, which leads to rapid solidification of the melt pool and disrupts the Marangoni convection, causing instability in the energy distribution of the high-temperature zone and resulting in significant drift of the geometric center. Excessive offset may lead to interlayer misalignment, necessitating prioritized monitoring in actual manufacturing.

Additionally, MP area and HT length were categorized as high sensitivity features. As scanning speed increases, the reduction in energy input amplifies the fluctuations in melt pool dimensions. Notably, the thermal flow distribution in the core region is more sensitive to scanning speed, prone to intermittent melting. However, due to constraints imposed by the laser spot size, the CV value changes in the high-temperature zone were not significant, and its width stability remained relatively high. This study, through systematic analysis of CV value changes, clarified the sensitivity classification of various melt pool characteristics to scanning speed, providing theoretical guidance for optimizing L-PBF parameters.

4. Conclusion

This study addresses the limitations of traditional coaxial melt pool monitoring in L-PBF, where spontaneous emission imaging often fails to capture low-radiation regions (such as the trailing edge), and conventional image processing methods lack robustness under complex backgrounds caused by auxiliary lighting, making real-time and accurate segmentation challenging. To overcome these issues, a new coaxial melt pool monitoring system was developed, along with an attention-enhanced deep learning segmentation network, Gcs-Unet. Experimental results demonstrate that the proposed system effectively captures detailed geometric and textural information of the melt pool, providing a solid foundation for precise online feature analytics. Gcs-Unet exhibited excellent performance under challenging conditions, achieving a global accuracy of 99.5 %, a Dice coefficient of 87.6 %, and an mIoU of 86.2 %, while reducing parameter count by 42.91 % compared to the baseline and achieving a per-frame inference time of 6.75 ms, significantly enhancing the potential for real-time deployment. In feature quantification, the relative errors of whole melt pool area and high-temperature zone area were 6.92 % and 13.88 %, respectively, demonstrating reliable and stable feature extraction capabilities.

This study also conducted a focused comparative analysis of melt pool feature parameters under different scanning speeds. The results indicated that the geometric center of the high-temperature zone was highly sensitive to scanning speed variation, with the coefficient of variation increasing by 37.45 % when the scanning speed was increased by 33 %. These findings provide useful theoretical insights and technical references for understanding melt pool dynamics and guiding process parameter optimization in L-PBF systems.

Despite the significant progress made, this study still has some limitations. For instance, the performance of the current method under certain extreme conditions requires further validation, and the applicability of the model under more complex process parameters needs additional investigation. Future work will focus on enhancing the model's generalization capability under diverse process settings, and expanding the range of studied parameters. In addition, further research could investigate the influence of different material systems on melt pool behavior, and explore the correlation between melt pool dynamics and final defect formation (e.g., porosity or lack of fusion), enabling a more comprehensive understanding of the process–structure–property relationship in L-PBF.

CRediT authorship contribution statement

Wei Wei: Writing – review & editing, Writing – original draft. **Yi Li:** Validation, Methodology. **Haixin Wu:** Validation. **Xiuming Li:** Visualization. **Yuhui Zhang:** Supervision. **Hang Ren:** Investigation. **Yu Long:** Writing – review & editing, Supervision. **Yunfei Huang:** Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests or personal relationships which may be considered as potential competing interests: Author Xiuming Li is currently employed by Guangxi Research Institute of Mechanical Industry co.,ltd. Other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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